
ASSESSMENT TOOL 1 – APPLICATION BASED QUESTIONS

NAME :- Ch Rama Karthik REDDY

REG NO :- 192472281

COURSE CODE :- DSA0305

COURSE NAME :- NATURAL LANGUAGE PROCESSING

1.Design and implement a Python-based resume information extraction system using Regular Expressions that performs the following tasks:

Extract the candidate's name.

Identify email addresses.

Extract mobile numbers.

Detect technical skills (Python, Java, SQL, Machine Learning, NLP).

Extract years of experience.

Generate a structured summary of the candidate's profile.

Display candidates who satisfy the minimum eligibility criteria (minimum 2 years of experience and Python skill).

```
import re
```

```
# Sample resumes
```

```
resumes = [
```

```
    """
```

```
Name: John Smith
```

```
Email: johnsmith@gmail.com
```

```
Mobile: 9876543210
```

```
Skills: Python, Java, SQL, Machine Learning
```

```
Experience: 3 years
```

```
    """,
```

```
    """
```

```
Name: Priya Sharma
```

Email: priya@gmail.com

Phone: 9123456789

Skills: Java, SQL

Experience: 1 year

""",

"""

Name: Rahul Kumar

Email: rahul@yahoo.com

Mobile: 9988776655

Skills: Python, NLP, Machine Learning

Experience: 5 years

"""

]

```
skills_list = ["Python", "Java", "SQL", "Machine Learning", "NLP"]
```

```
print("===== Resume Information Extraction =====\n")
```

```
for resume in resumes:
```

```
    # Extract Name
```

```
    name = re.search(r"Name:\s*(.*)", resume)
```

```
    name = name.group(1) if name else "Not Found"
```

```
    # Extract Email
```

```
    email = re.search(r"[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}", resume)
```

```
    email = email.group() if email else "Not Found"
```

```
    # Extract Mobile Number
```

```
    mobile = re.search(r"\b\d{10}\b", resume)
```

```
    mobile = mobile.group() if mobile else "Not Found"
```

```
# Extract Skills
found_skills = []
for skill in skills_list:
    if re.search(skill, resume, re.IGNORECASE):
        found_skills.append(skill)

# Extract Experience
exp = re.search(r"(\d+)\s+year", resume, re.IGNORECASE)
experience = int(exp.group(1)) if exp else 0

# Structured Summary
print("Candidate Profile")
print("-----")
print("Name      :", name)
print("Email     :", email)
print("Mobile    :", mobile)
print("Skills    :", ", ".join(found_skills))
print("Experience :", experience, "years")

# Eligibility Check
if experience >= 2 and "Python" in found_skills:
    print("Status   : Eligible")
else:
    print("Status   : Not Eligible")

print()
```

Output:

```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/Siva Polam/Downloads/section 1question.py =====
===== Resume Information Extraction =====

Candidate Profile
-----
Name       : John Smith
Email      : johnsmith@gmail.com
Mobile     : 9876543210
Skills     : Python, Java, SQL, Machine Learning
Experience : 3 years
Status     : Eligible

Candidate Profile
-----
Name       : Priya Sharma
Email      : priya@gmail.com
Mobile     : 9123456789
Skills     : Java, SQL
Experience : 1 years
Status     : Not Eligible

Candidate Profile
-----
Name       : Rahul Kumar
Email      : rahul@yahoo.com
Mobile     : 9988776655
Skills     : Python, Machine Learning, NLP
Experience : 5 years
Status     : Eligible
>>>
```

Ln: 34 Col: 0

2. An e-commerce company plans to enhance its product search engine to improve customer experience through flexible keyword matching.

Design and implement a Python-based product search system using Regular Expressions that performs the following tasks:

Search products using an exact keyword.

Perform prefix-based searches.

Perform suffix-based searches.

Search using partial keywords.

Perform case-insensitive searches.

Display all matching product names.

Generate a report showing the total number of matching products for each search.

```
import re
```

```
# Product List
```

```
products = [  
    "Apple iPhone 15",  
    "Apple Watch",  
    "Samsung Galaxy S24",  
    "Sony Headphones",  
    "Dell Laptop",  
    "HP Laptop",  
    "Lenovo Laptop",  
    "Apple AirPods",  
    "Bluetooth Speaker",  
    "Wireless Mouse"
```

```
]
```

```
# Function for Product Search
```

```
def search_products(pattern, flag=0):
```

```
    result = [p for p in products if re.search(pattern, p, flag)]
```

```

for p in result:
    print(p)
print("Total Matching Products:", len(result))
print()

print("----- Exact Keyword Search (Laptop) -----")
search_products(r"\bLaptop\b")

print("----- Prefix Search (Apple) -----")
search_products(r"^Apple")

print("----- Suffix Search (Laptop) -----")
search_products(r"Laptop$")

print("----- Partial Keyword Search (Phone) -----")
search_products(r"Phone")

print("----- Case-Insensitive Search (apple) -----")
search_products(r"apple", re.IGNORECASE)

```

Output:

```

Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
= RESTART: C:/Users/Siva Polam/AppData/Local/Programs/Python/Python313/at1 secti
on 2 question.py
----- Exact Keyword Search (Laptop) -----
Dell Laptop
HP Laptop
Lenovo Laptop
Total Matching Products: 3

----- Prefix Search (Apple) -----
Apple iPhone 15
Apple Watch
Apple AirPods
Total Matching Products: 3

----- Suffix Search (Laptop) -----
Dell Laptop
HP Laptop
Lenovo Laptop
Total Matching Products: 3

----- Partial Keyword Search (Phone) -----
Apple iPhone 15
Total Matching Products: 1

----- Case-Insensitive Search (apple) -----
Apple iPhone 15
Apple Watch
Apple AirPods
Total Matching Products: 3

>>>

```

3.A university is developing an automated online registration portal to verify student information before course enrollment.

Design and implement a Python-based validation system using Regular Expressions that performs the following tasks:

Validate the student register number.

Validate the institutional email address.

Validate the course code.

Validate the semester information.

Validate the student's mobile number.

Display appropriate validation messages for each field.

Generate a final registration status report indicating whether the registration is successful.

```
import re

# Student Details
reg_no = "23CS001"
email = "student@saveetha.edu.in"
course = "CS301"
semester = "5"
mobile = "9876543210"

status = True

# Register Number
if re.fullmatch(r"\d{2}[A-Z]{2}\d{3}", reg_no):
    print("Register Number : Valid")
else:
    print("Register Number : Invalid")
    status = False

# Institutional Email
if re.fullmatch(r"[a-zA-Z0-9._%+-]+@saveetha\.edu\.in", email):
    print("Email      : Valid")
else:
    print("Email      : Invalid")
    status = False

# Course Code
if re.fullmatch(r"[A-Z]{2}\d{3}", course):
    print("Course Code   : Valid")
else:
    print("Course Code   : Invalid")
    status = False

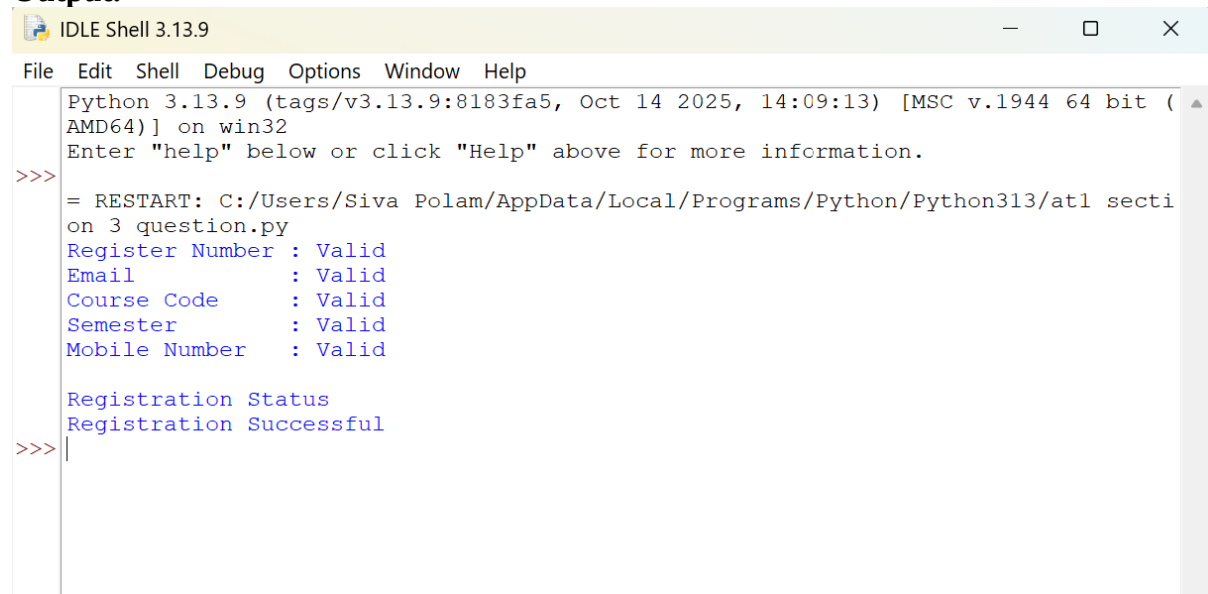
# Semester
```

```
if re.fullmatch(r"[1-8]", semester):
    print("Semester      : Valid")
else:
    print("Semester      : Invalid")
    status = False

# Mobile Number
if re.fullmatch(r"[6-9]\d{9}", mobile):
    print("Mobile Number  : Valid")
else:
    print("Mobile Number  : Invalid")
    status = False

# Final Registration Status
print("\nRegistration Status")
if status:
    print("Registration Successful")
else:
    print("Registration Failed")
```

Output:



```
IDLE Shell 3.13.9
File Edit Shell Debug Options Window Help
Python 3.13.9 (tags/v3.13.9:8183fa5, Oct 14 2025, 14:09:13) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/Siva Polam/AppData/Local/Programs/Python/Python313/at1 secti
on 3 question.py
Register Number : Valid
Email           : Valid
Course Code     : Valid
Semester        : Valid
Mobile Number   : Valid

Registration Status
Registration Successful
>>> |
```